

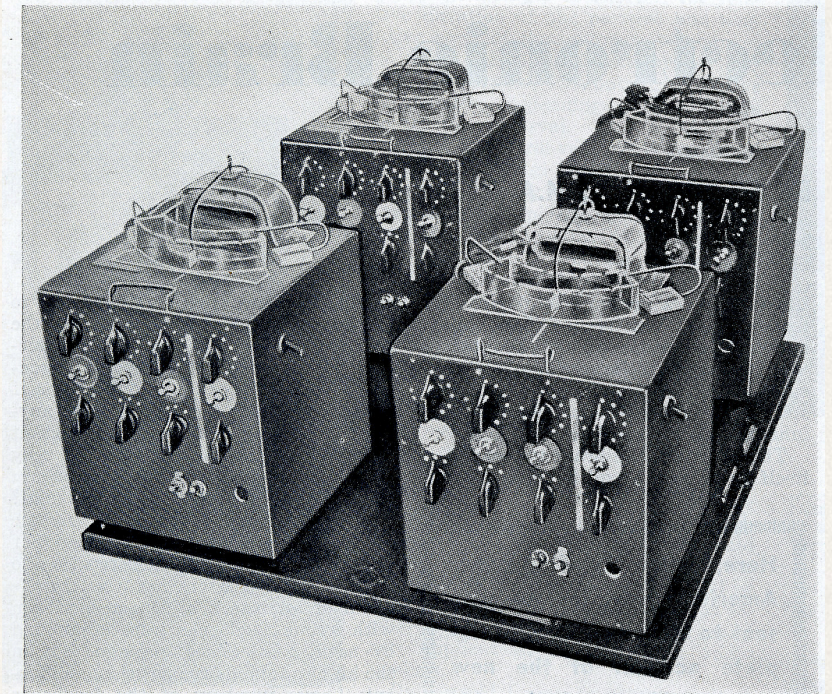
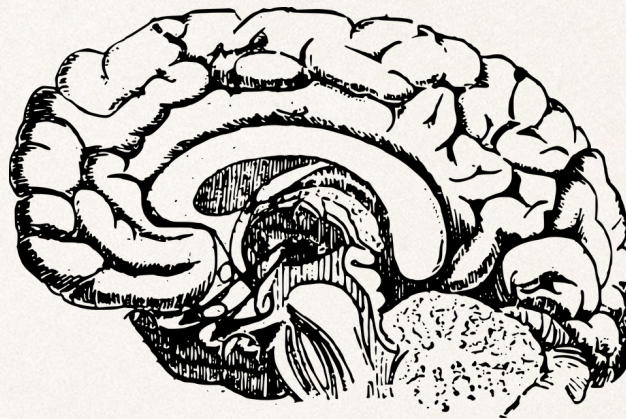
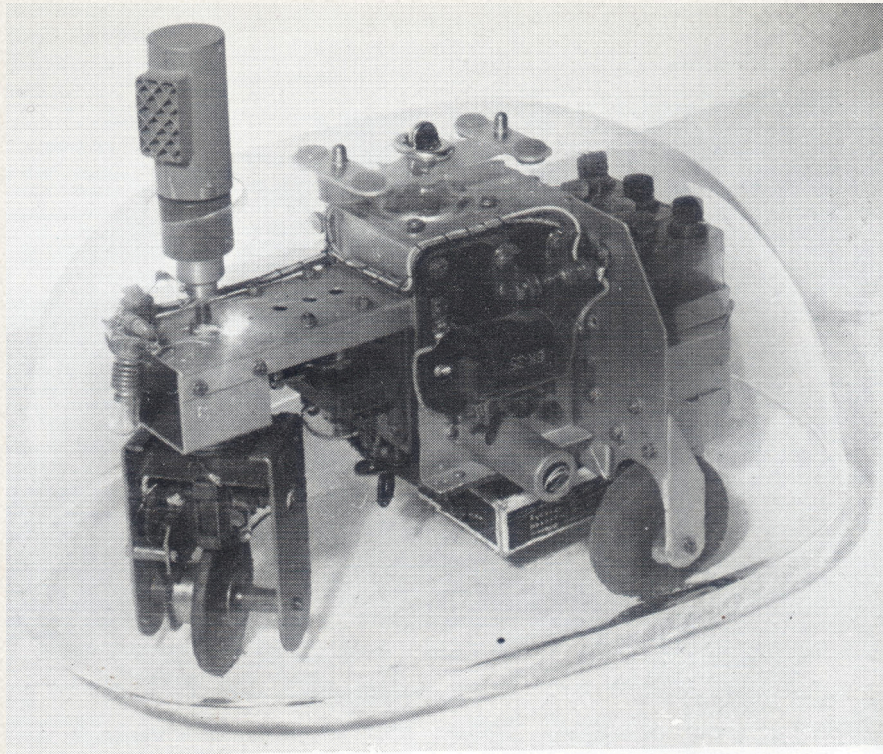


Fig. 1—The homeostat, with its four units, each one of which reacts on all the others.

Nonmodular Architectures of Cognitive Systems based on Active Inference

Manuel Baltieri

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- Visitor @ EON (ELSI Origins of life Network), ELSI, Tokyo Tech, Tokyo (Japan)

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Roadmap

- ❖ Modularity, especially in cognitive science
- ❖ Separation principle in control theory (and neuroscience)
- ❖ An alternative in *active inference*

Modularity?

Intuitive definition

Decomposability (and near-decomposability) - Herbert A. Simon

Structural and functional - Watson and Pollack (2005)

“The modular mind” - Jerry A. Fodor

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“The main theoretical findings from the approach can be summed up in two propositions:

(a) in a nearly decomposable system, the short run behavior of each of the component subsystems is approximately independent of the short-run behavior of the other components;

(b) in the long run, the behavior of any one of the components depends in only an aggregate way on the behavior of the other components. ”

– *“The Architecture of Complexity”. Herbert A. Simon*
(on near-decomposability)

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Fodorian modularity

Modules are:

- ❖ domain specific
- ❖ innately specified
- ❖ informationally encapsulated
- ❖ fast
- ❖ hardwired (neurally specific)
- ❖ autonomous
- ❖ not assembled

Fodorian modularity

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- ❖ **autonomous**
- ❖ **not assembled**

The sandwich of cognitive science



Perception

**Cognitivism/
computationalism**

Themes:

Feedforward computation,
(informationally) independent
components, symbol
manipulation, central
processing, input-output
structure, information
processing and encoding,
(internal) representations, etc.

Cognition

Action

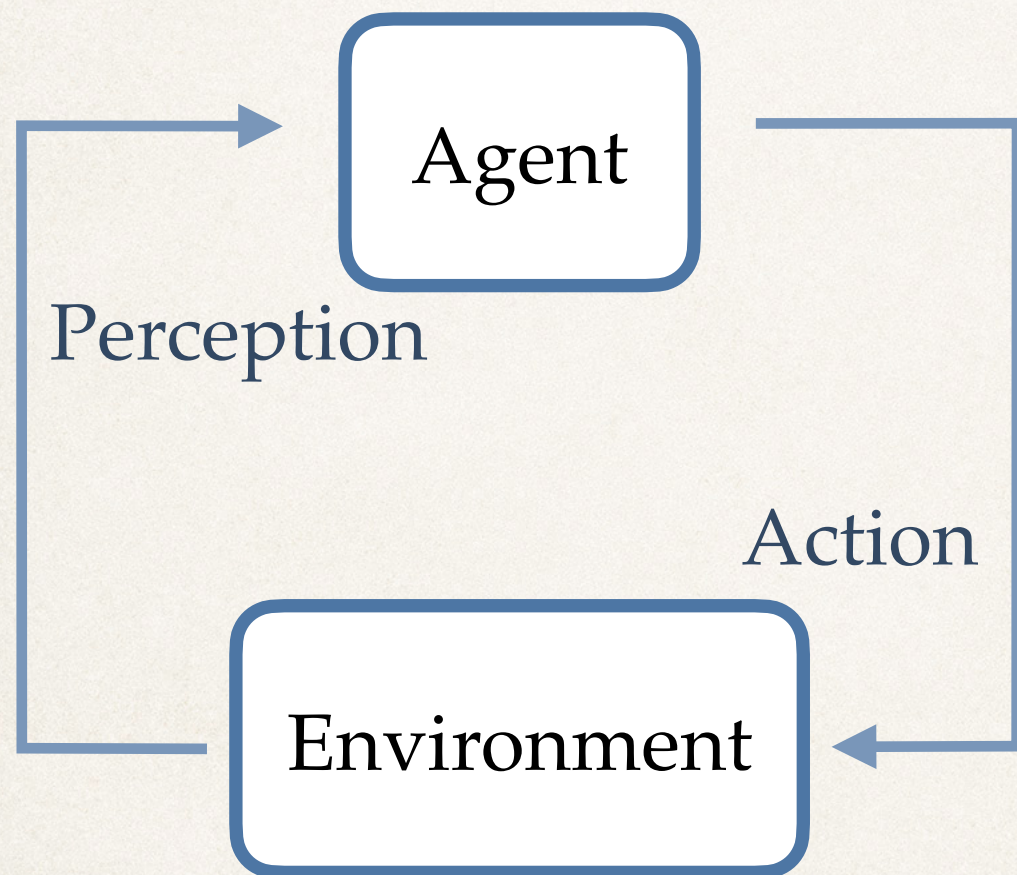
But the brain *is not* (just) a feedforward network of bottom-up signals

Feedback (top-down and lateral) connections are everywhere

The brain *is not* an isolated system

The brain resides in embodied and situated agents

The 4E views, agents and environments (and to an extent, RL)



4Es: Embodied, Enactive, Extended and Embedded

Themes:

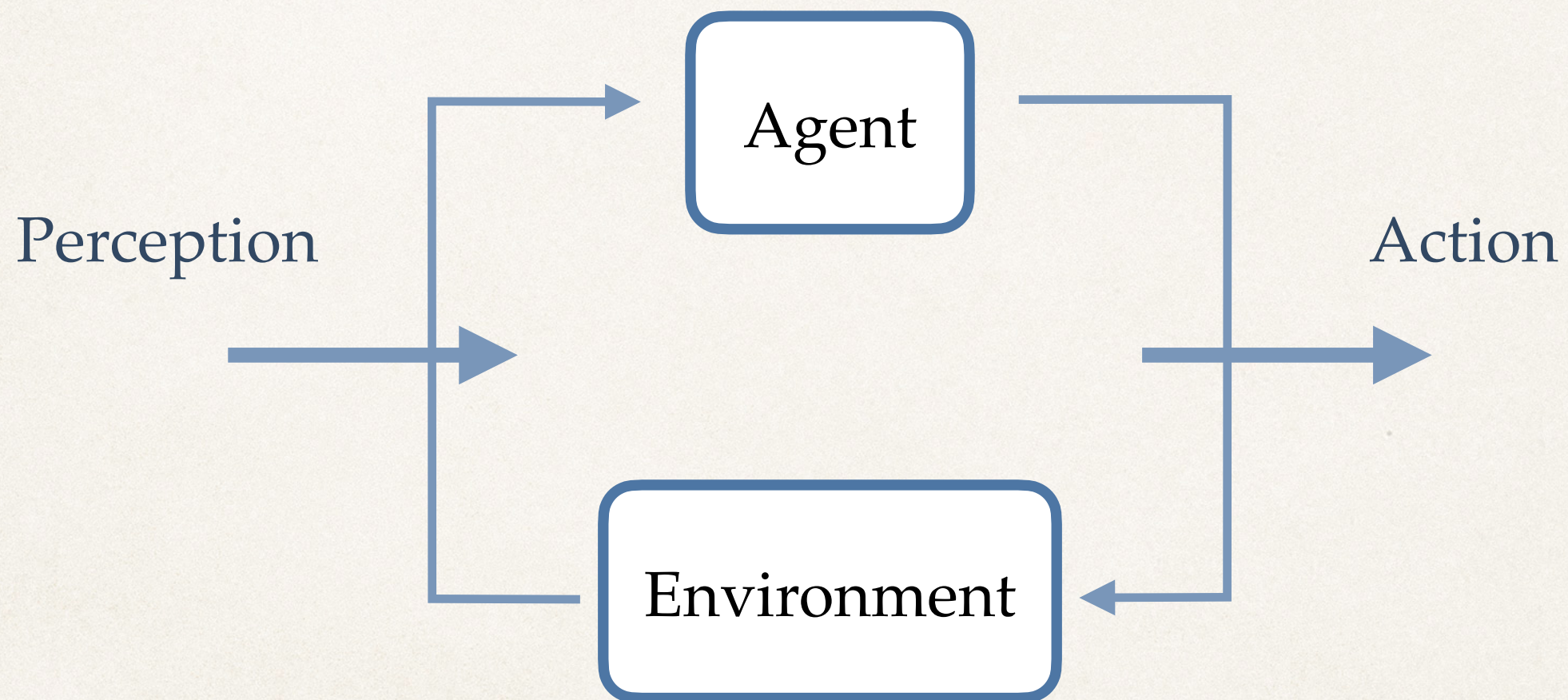
Feedback from environment, role of the body, situatedness, dynamics (time), autonomy, affordances, sensorimotor coupling, etc.

Examples:

- ❖ Cybernetics (Wiener, Ashby, ...),
- ❖ Behaviour-based robotics (Brooks),
- ❖ Evolutionary robotics (Harvey, Husbands, Floreano, Bongard, ...),
- ❖ Developmental robotics (Lungarella, ...),
- ❖ Dynamical systems approaches (Beer),
- ❖ etc.

We like modules...

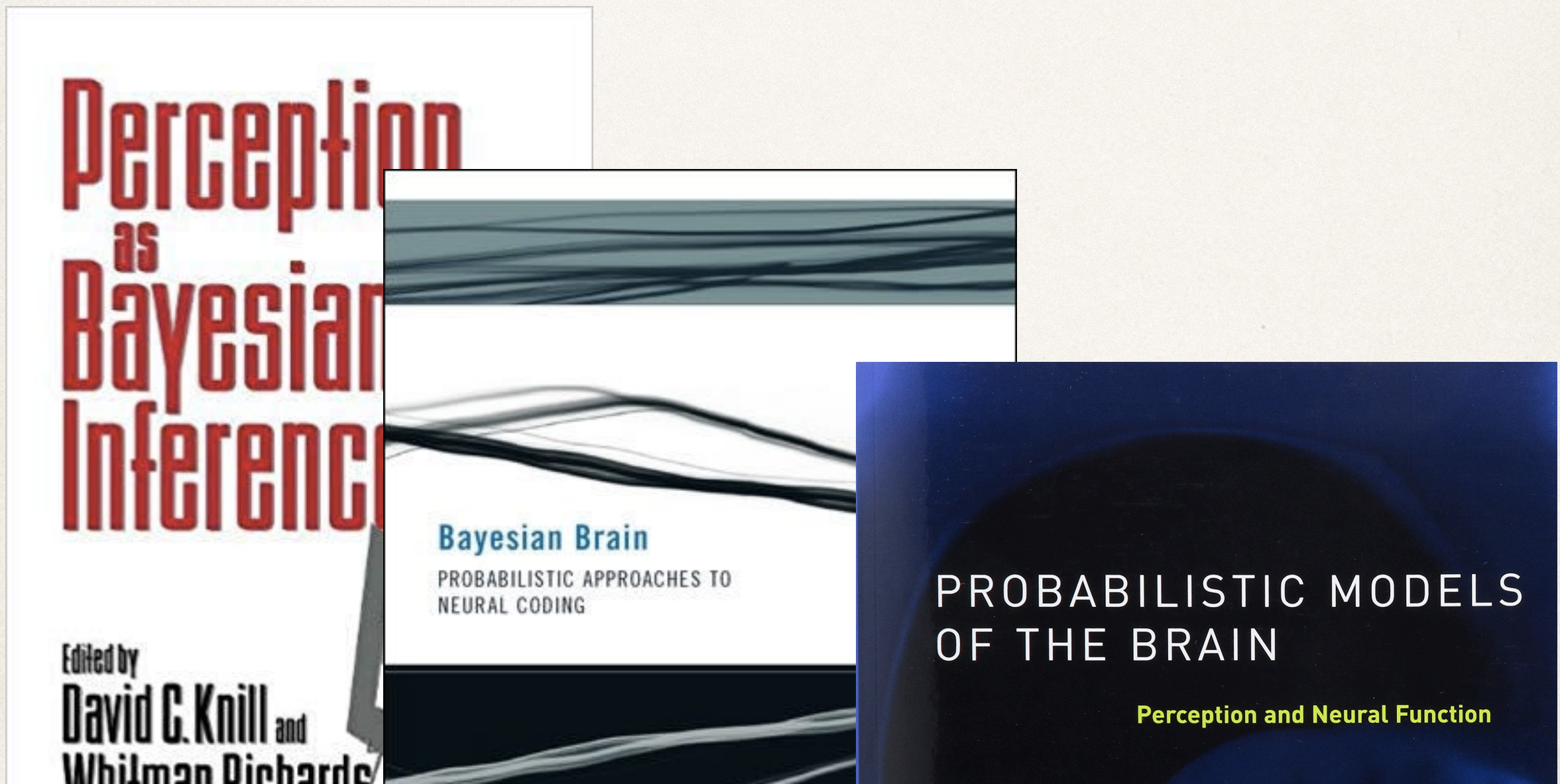
Input-output architectures (classical sandwich)



Meanwhile, in modern neuroscience...

Claim 1

Perception can be described as a process of (Bayesian) inference or estimation



Claim 2

Action can be described as a process of (optimal) control

718

Internal models for motor control and trajectory planning

Mitsuo Kawato

A number of internal model concepts are now being tested in neuroscience and cognitive science. These concepts are supported by behavioral, neurophysiological and computational data. Furthermore, these models have had their functions revealed by such data. In particular, the role of internal models on inverse dynamics model learning is directly supported by unit recordings from cerebellar Purkinje cells. In contrast, forward inverse models describing how diverse environments can be controlled and learned have not been proposed. The 'minimum variance' model has been another major recent advance in the computational study of motor control. This model integrates two fundamental approaches on trajectory planning, strongly supporting both kinematic and dynamic internal models for movement planning and control.

Addresses

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review

Computational principles of movement neuroscience

Daniel M. Wolpert¹ and Zoubin Ghahramani²

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nature
neuroscience

Unifying principles of movement have emerged from the computational approach to movement control. This review examines several of these principles and shows how they can be derived from the computational approach to movement control.

The computational study of motor control is fundamentally concerned with the relationship between sensory signals and motor commands. The transformation from motor commands to sensory consequences is governed by the physics of movement, the musculoskeletal system and sensory processing.

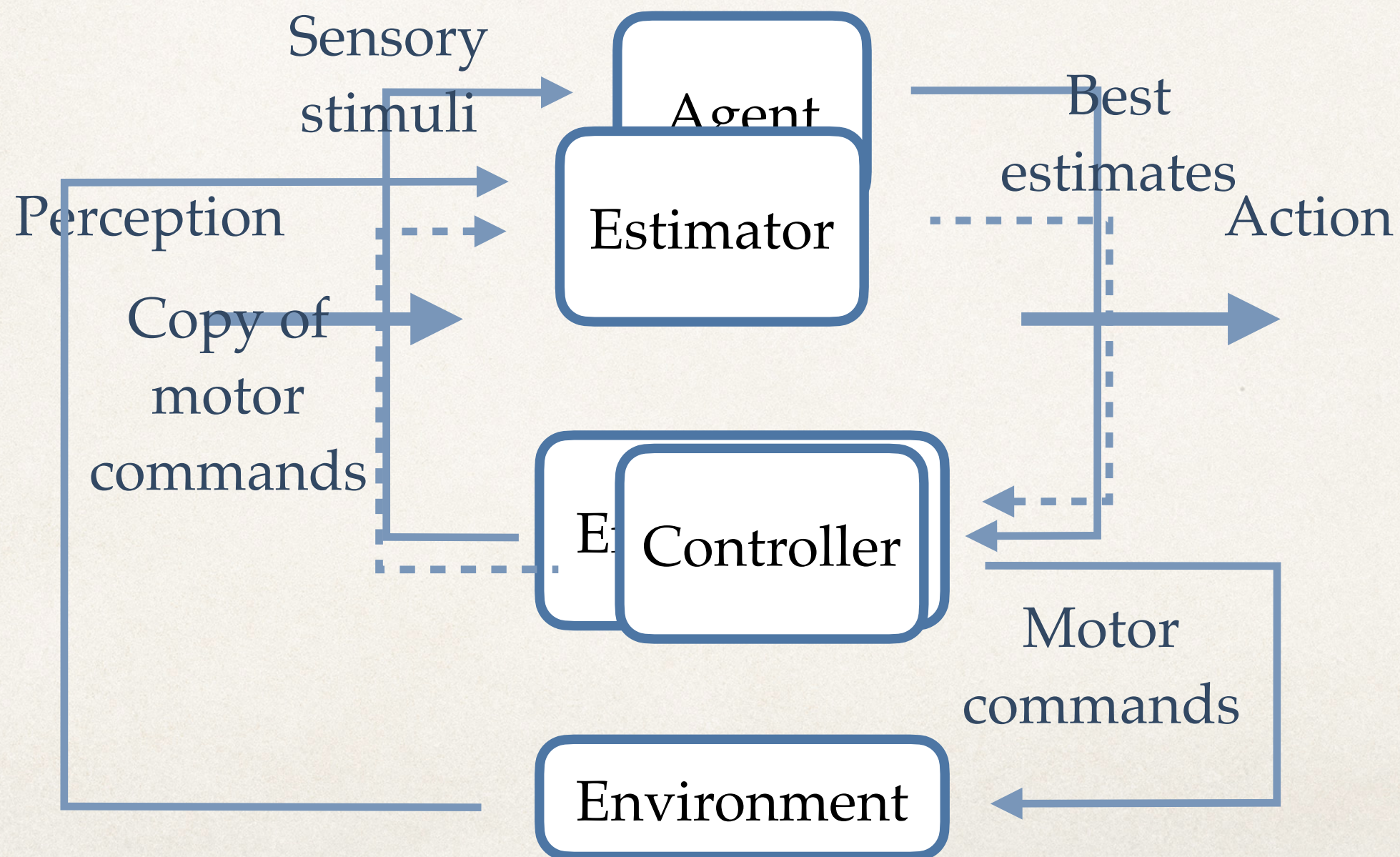
Optimality principles in sensorimotor control

Emanuel Todorov

REVIEW

We really like modules...

Perception and action machines (classical and modern) science



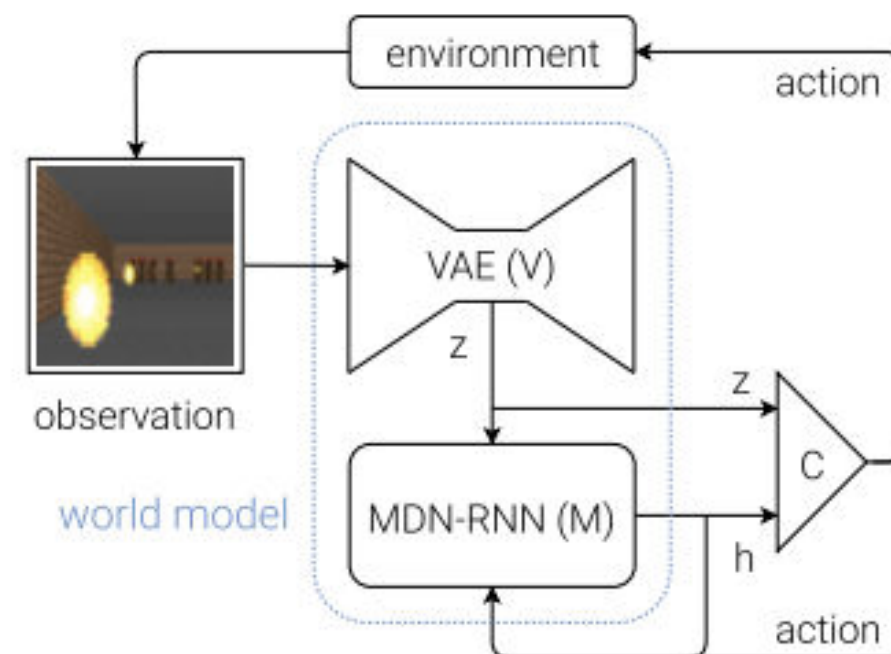
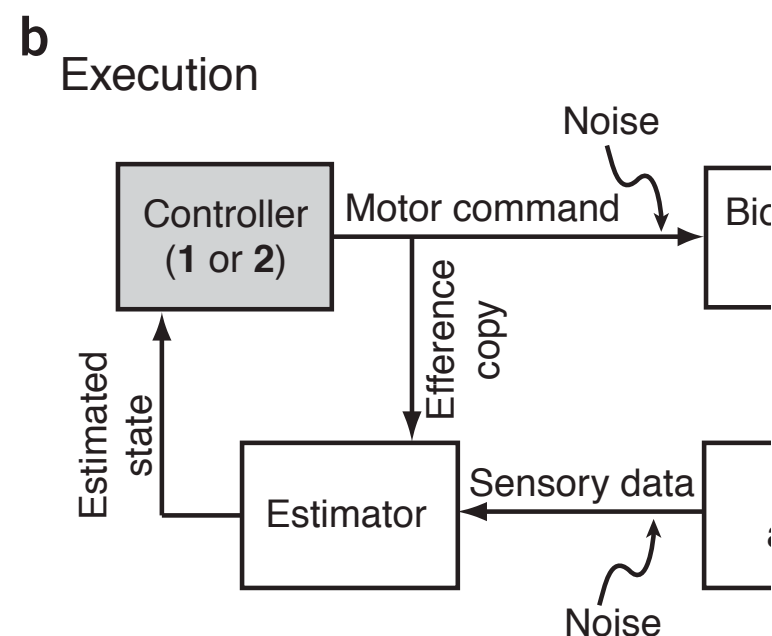
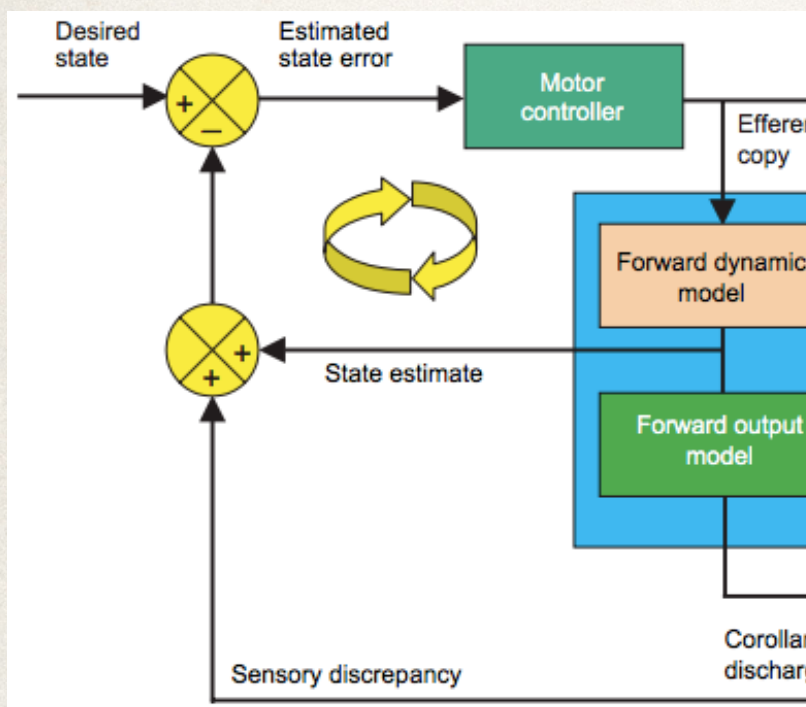
Claim 3 (often implicit)

The separation principle

Classic result in control theory (cf. “certainty equivalence” in econometrics and separation principle in information theory) for **linear systems**:

LQG (Linear Quadratic Gaussian) control =

Kalman filter (**estimator**) + Linear quadratic regulator (**controller**)



Wolpert et al. (1998)

Todorov (2004)

Ha et al. (2018), but this one is “ok”

Modular minds and the separation principle

*Fodor (modularity)
classical sandwich*

Perception



Cognition



Action

Cog. (neuro)science

Estimation / inference



(Complicated stuff or
"just inference", à la Friston)



Optimal control

*Control theory,
separation principle*

Kalman(-Bucy)
filter



(Complicated stuff)



Linear Quadratic
Regulator

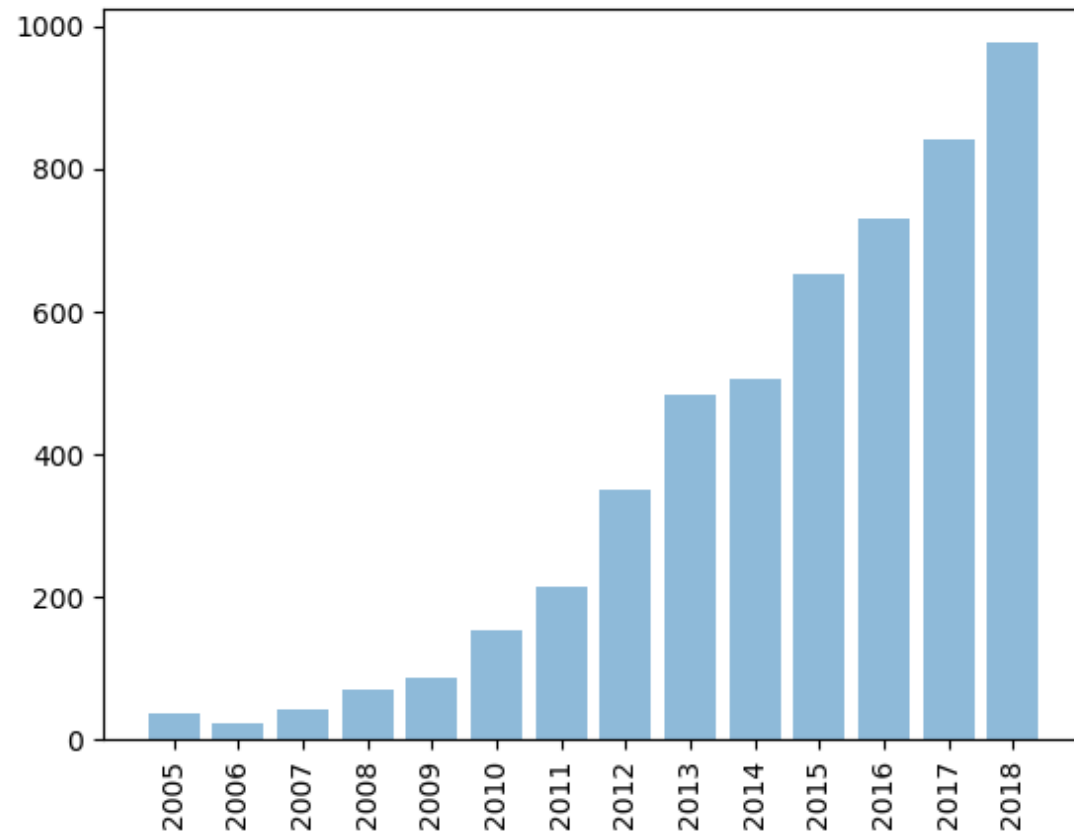
The free energy principle and active inference



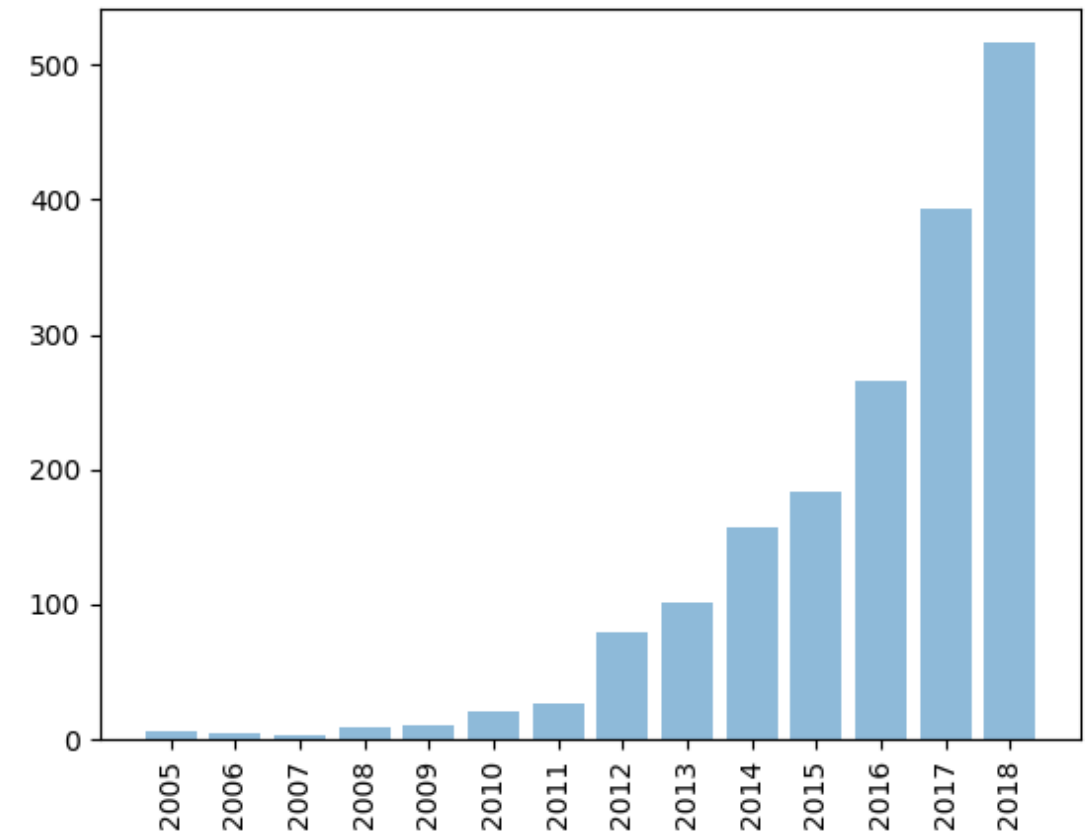
Karl Friston (nice profile at <https://www.wired.com/story/karl-friston-free-energy-principle-artificial-intelligence/>)

- ❖ Psychiatrist with a background in physical sciences
- ❖ Creator of SPM, VBM and DCM (statistical methods for the analysis of neuroimaging data)
- ❖ > 200000 citations, h-index > 200

Google scholar: "free energy" Friston



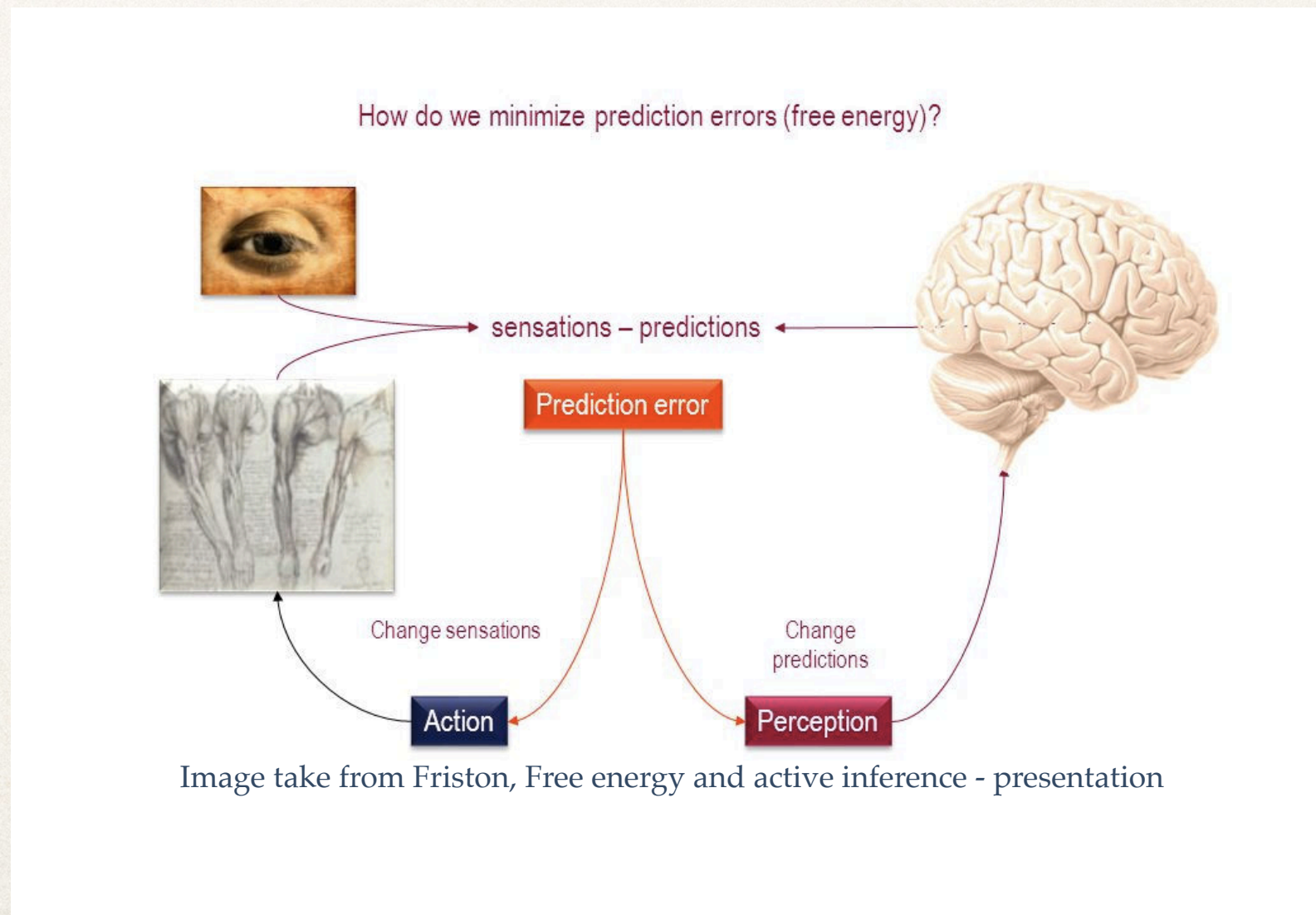
Google scholar: "active inference" Friston



The free energy principle

Hypotheses: Perception and action can be described as processes of (prediction) error minimisation via Bayesian inference
(claims 1 and 2, but not 3)

Active
inference



Perceptual
inference

The variational free energy formulation

$$F \approx -\ln p(y, \vartheta) \Big|_{\vartheta=\mu_\vartheta} \quad \vartheta = \{x, v, \theta, \gamma\}$$

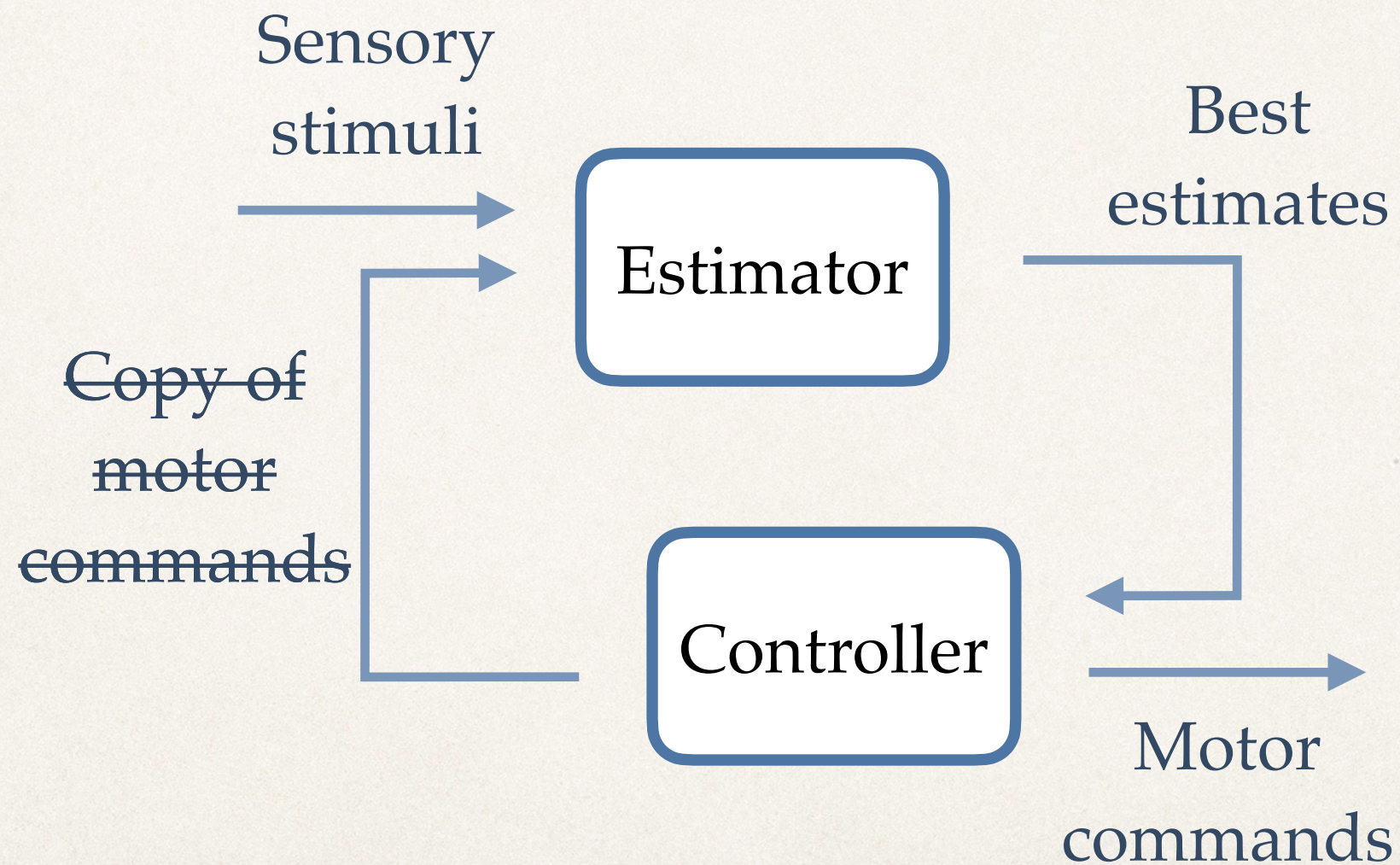
❖ Perception

$$\dot{\mu}_x = D\mu_x - \frac{\partial F}{\partial \mu_x}$$

❖ Action

$$\dot{a} = \frac{\partial F}{\partial a} = \frac{\partial F}{\partial y} \frac{\partial y}{\partial a}$$

What about active inference?



(Exteroception)

Sensory
stimuli

Best
estimates

(Interoception)

Sensory
stimuli

Best
estimates

(Proprioception)

Sensory
stimuli

Best
estimates

Reflex arc

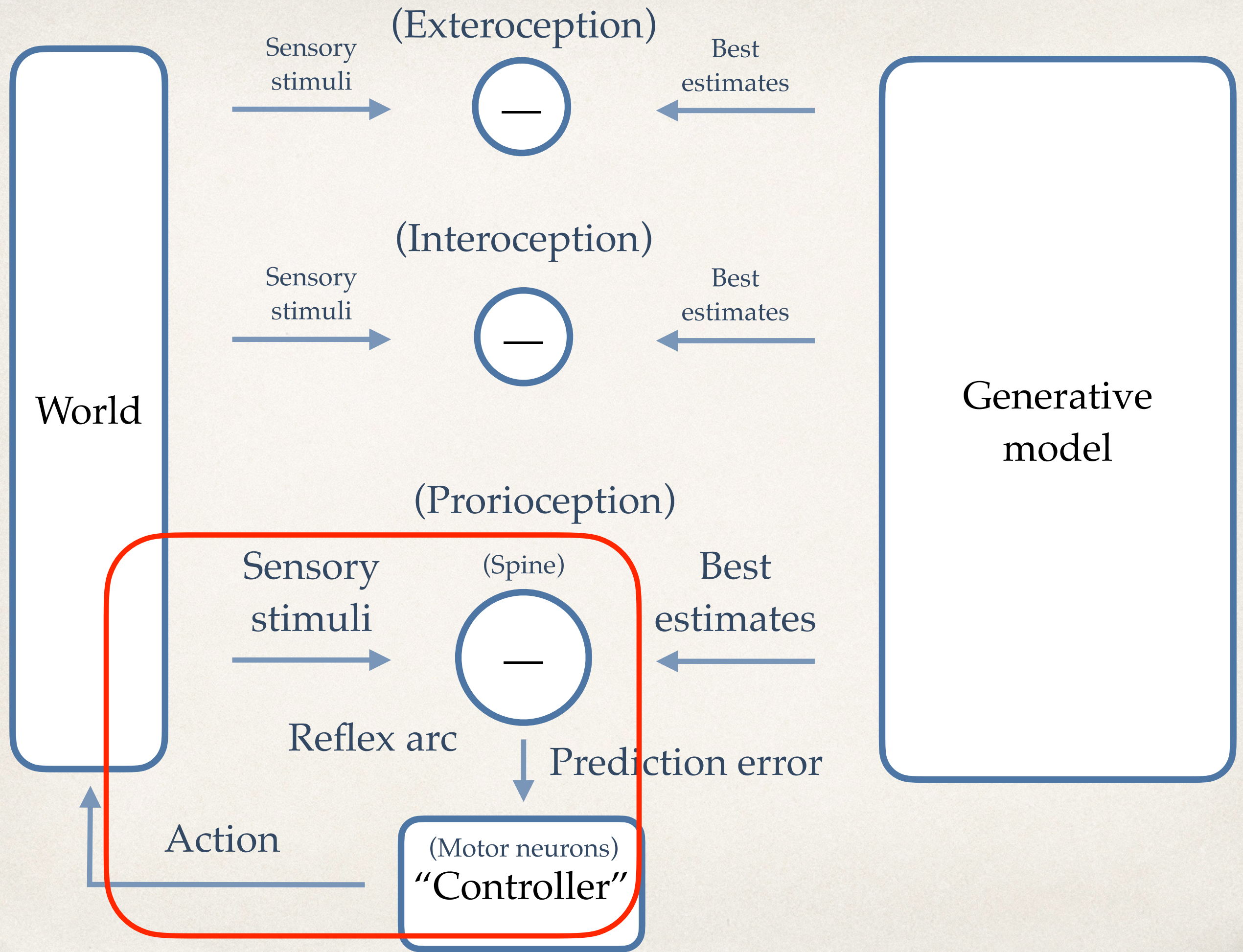
Prediction error

Action

(Motor neurons)
"Controller"

World

Generative
model

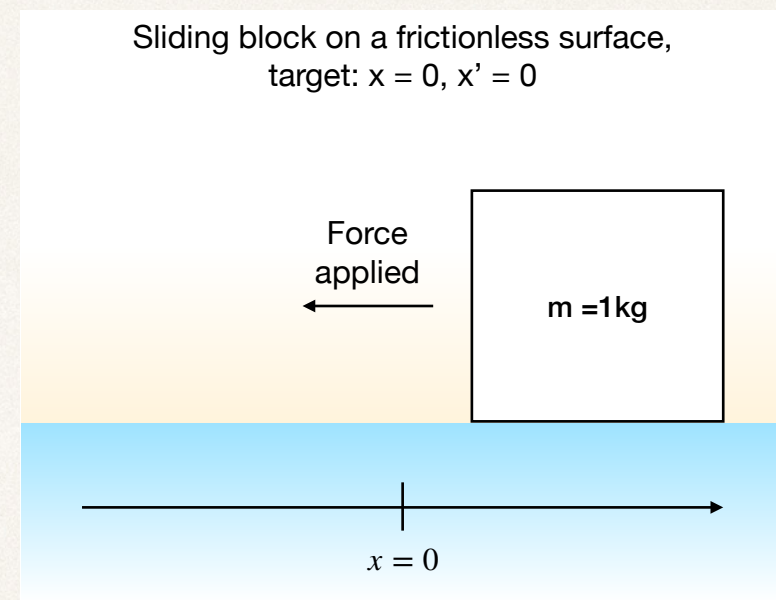


LQG vs active inference

Minimal model reproducing LQG vs active inference

Comparison of LQG (separation principle) and active inference equations

The role of efference copy and external forces + implications for models of behaviour



where matrix

$$A =$$

and covarian

$$\Sigma_z =$$



+ z

Model of
a single-joint
system

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & \exp(-1) \end{bmatrix}$$

LQG vs active inference

Minimal model reproducing LQG
vs active inference

**Comparison of LQG (separation
principle) and active inference
equations**

The role of efference copy and
external forces + implications for
models of behaviour

LQG

$$\dot{\hat{x}} = A\hat{x} + Ba + K(y - C\hat{x})$$

$$a = -L\hat{x}$$

$$K = PH^T(\Sigma_z)^{-1}$$

$$L = R^{-1}B^TV$$

$$\dot{P} = \Sigma_w + AP + PA^T - K(\Sigma_z)K^T$$

$$-\dot{V} = Q + A^TV + VA - L^TRL$$

Active inference

$$F \approx \frac{1}{2} \left[\left(\mathbf{y} - \hat{C}\boldsymbol{\mu}_x \right)^T \Pi_z \left(\mathbf{y} - \hat{C}\boldsymbol{\mu}_x \right) + \right. \\ \left. + \left(\boldsymbol{\mu}'_x - \hat{A}\hat{\boldsymbol{\mu}}_x - \hat{B}\boldsymbol{\mu}_v \right)^T \Pi_w \left(\boldsymbol{\mu}'_x - \hat{A}\boldsymbol{\mu}_x - \hat{B}\boldsymbol{\mu}_v \right) + \right. \\ \left. - \ln |\Pi_z| - \ln |\Pi_w| + (m + n) \ln 2\pi \right]$$

$$\dot{\boldsymbol{\mu}}_x = D\boldsymbol{\mu}_x - \frac{\partial F}{\partial \boldsymbol{\mu}_x} = \boldsymbol{\mu}'_x + \hat{C}^T \Pi_z \left(\mathbf{y} - \hat{C}\boldsymbol{\mu}_x \right) + \\ + \hat{A}^T \Pi_w \left(\boldsymbol{\mu}'_x - \hat{A}\boldsymbol{\mu}_x - \hat{B}\boldsymbol{\mu}_v \right)$$

$$\dot{\boldsymbol{\mu}}'_x = D\boldsymbol{\mu}'_x - \frac{\partial F}{\partial \boldsymbol{\mu}'_x} = \boldsymbol{\mu}''_x - \Pi_w \left(\boldsymbol{\mu}'_x - \hat{A}\boldsymbol{\mu}_x - \hat{B}\boldsymbol{\mu}_v \right)$$

$$\dot{a} = -\frac{\partial F}{\partial a} = -\frac{\partial F}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial a} = -\frac{\partial \mathbf{y}^T}{\partial a} \Pi_z \left(\mathbf{y} - \hat{C}\boldsymbol{\mu}_x \right)$$

The actual differences

Standard formulation (separation principle, LQG)

$$\begin{aligned}\dot{x} &= Ax + Bu + w \\ y &= Cx + z\end{aligned}$$

u : external, or exogenous, inputs

What are the inputs?

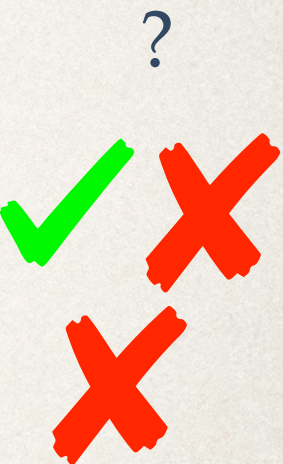
$$\begin{aligned}\dot{x} &= Ax + Ba + I + w \\ y &= Cx + z\end{aligned}$$

$$u = a + I$$

a : a copy of motor actions,
efferent copy

I : (actual) *external* inputs

v : priors, *desired trajectories* for movement



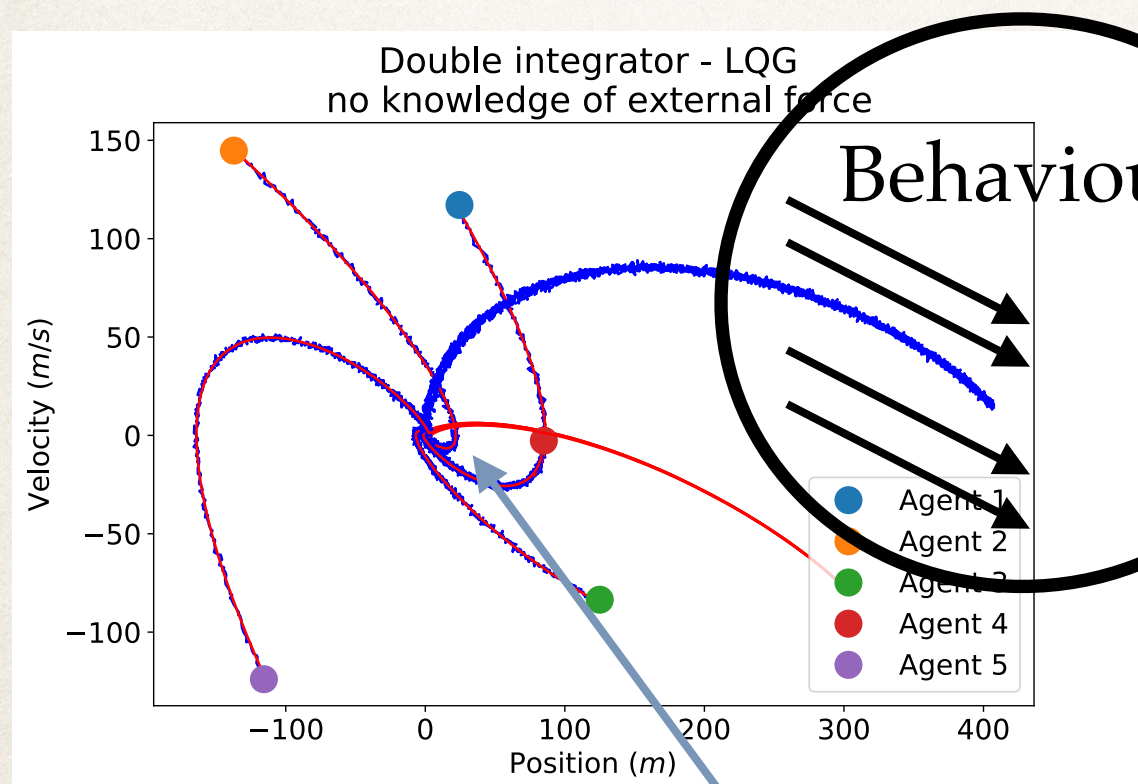
LQG vs active inference

Minimal model reproducing LQG
vs active inference

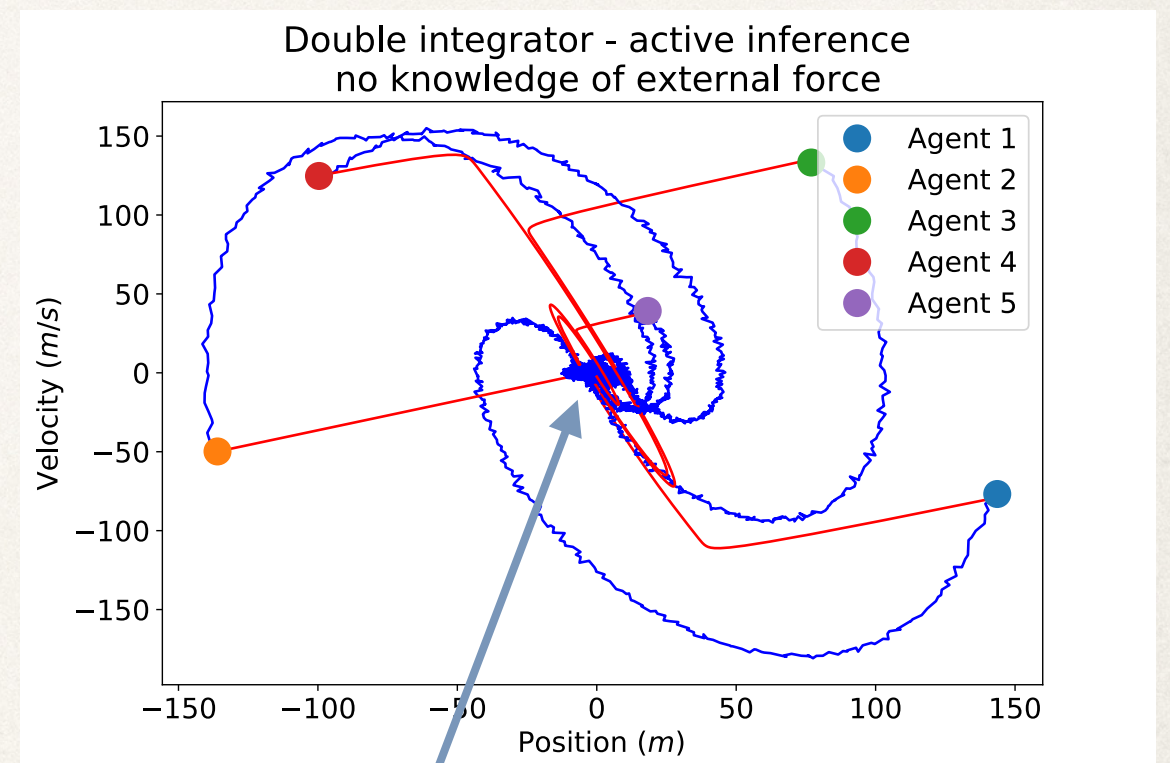
Comparison of LQG (separation
principle) and active inference
equations

**The role of efference copy and
external forces + implications
for models of behaviour**

LQG vs active inference, with external disturbance (e.g., wind)



Target



Target

Summary

- ❖ We like modularity (biology, engineering, software production, economics, etc.), and modularity is good
- ❖ Modularity in cognitive science and neuroscience is a standard assumption (classical sandwich, input-output architectures), but not very realistic for models of cognition and behaviour
- ❖ Modularity aligns well (?) with the separation principle of control theory
- ❖ The separation principle is only but a linear approximation of feedback systems and copies of motor signals may not be biological plausible
- ❖ Active inference drops the separation assumption and applies Bayesian inference and optimal control to more general problems (e.g., unknown forces that affect observations) with no need for classical modularity

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